The Router Is the Demand Curve: Price Steering and Delayed Credit in AI Inference Markets

Anonymous Authors

Abstract

Open-weight inference is a partially substitutable execution service sold by providers through routers. We study a documented inverse-square allocation rule and a conditional selection association for recent price cutters. Inverse-power routing creates a classical markup floor. More importantly, a finite cut penalty creates a delayed-credit region: cutting is immediately unprofitable yet optimal after the penalty expires. In a calibrated two-price reduction, the rational boundary is 9.24 periods. At the seven-period calibration, exact optimization cuts while primitive-action Q-learning remains high in 19 of 20 seeds. A path-equivalent commitment option preserves the optimal value, restores the cut in 18 of 20 seeds, and lowers normalized regret by 6.43 percentage points (paired seed-bootstrap 95% interval $[4.93, 7.55]$). The regret effect remains beneficial in all nine registered Q-learning parameter cells. Across four calibrated price books, its sign aligns with the rational memory boundary, although the strict transport gate fails. An eight-step TD target does not improve on one-step Q, narrowing the result to option-specific temporal abstraction. Thus router memory can separate rational incentives from learned pricing, but interventions are nonmonotone. This is a calibrated counterfactual mechanism, not evidence of provider conduct or live-router causality.

1 Introduction

Open weights separate a model's production technology from the firm that executes it. Multiple providers can serve the same model, while a router filters eligible providers and dispatches each request. The traded object is partially commodity-like: model identity is common, but sampling, tools, serving stacks, latency, and reliability prevent perfect substitutability. The router—rather than the end user—therefore determines the residual demand faced by each provider.

This paper studies a simple but consequential design: after eligibility and uptime filtering, non-tool-calling default selection is documented as price-weighted with probability proportional to inverse squared price [OpenRouter, 2026]. A public rule of this form directly maps posted price into route share. Its exponent determines the static elasticity faced by providers. A second, history-dependent design margin changes the allocation weight of recent price cutters. Such a rule may be a quality safeguard against bait-and-switch pricing, yet it also delays the benefit from a persistent cut.

We separate three questions that are often conflated in algorithmic-pricing work. First, what static pricing incentives does the routing rule create? Second, when does a history-dependent penalty make a persistent cut rational but difficult for a bounded learner to discover? Third, can an interface intervention repair that learning failure without expanding the provider's feasible price paths? We do not infer collusion from elevated prices, and we do not treat a learned policy as an equilibrium without a deviation audit.

Our contributions are:

1. We map inverse-power allocation to a constant-elasticity logit demand system. The documented exponent $a = 2$ places symmetric duopoly at a fixed-demand knife edge and creates a provider-level markup floor that free entry cannot eliminate.
2. We derive an exact memory boundary for a cut penalty. In the delayed-credit region, a persistent cut is optimal even though every penalized cut period pays less than staying high.
3. We construct a deterministic finite-state laboratory from the audited high quote and best persistent cut. Full state observability does not solve the learning failure. A semi-Markov commitment option does, while preserving the exact optimal value and every feasible primitive path. A registered eight-step TD falsification does not reproduce the improvement.
4. We retain the negative results. The original high-price equilibrium interpretation fails a multi-period deviation audit; the state-aliasing hypothesis fails; and the strict cross-market transport gate fails. The supported claim is the narrower delayed-credit mechanism.

2 Market and static incentives

One market is a model–workload pair. Provider i posts $p_i \in (0, \overline{p}]$, has marginal cost c_i, and faces a unit mass D of requests. The router assigns first-route share

$$s_i(p) = \frac{p_i^{-a}}{\sum_j p_j^{-a}}, \qquad a > 0, \tag{1}$$

and provider profit is $\pi_i = D s_i(p)(p_i - c_i)$. The baseline treats D as fixed; elastic demand is a sensitivity, not an estimated structural model.

Lemma 1 (Share elasticity and single crossing)**.** *For fixed rival prices,*

$$\frac{\partial s_i}{\partial p_i} = -\frac{a}{p_i} s_i(1 - s_i).$$

Profit is strictly quasiconcave above cost. Its unique best response is the root of

$$a(1 - s_i(p_i))\frac{p_i - c_i}{p_i} = 1$$

when that root lies below $\overline{p}$, and is $\overline{p}$ otherwise.

Theorem 2 (Symmetric equilibrium and markup floor)**.** *Suppose costs are symmetric at c.*

(i) If $a(n-1) > n$ and the cap does not bind, the unique symmetric interior equilibrium is

$$p^* = c\frac{a(n-1)}{a(n-1) - n}.$$

(ii) If $a(n-1) \le n$, the only symmetric equilibrium is the menu ceiling. At $a = 2$, this is exactly the duopoly boundary.

(iii) At any interior equilibrium, symmetric or not, and with any outside option added to the allocation denominator,

$$p_i > c_i\frac{a}{a-1} \qquad (a > 1).$$

The bound is tight as provider share vanishes.

The static structure is classical differentiated-products pricing; the contribution is the mapping of a published platform rule into the demand primitive. The router's smoothing makes same-model providers imperfect substitutes from the provider's perspective. The exponent is therefore a policy dial, not a taste coefficient estimated here.

For a sensitivity with $D = D_0 P^{\varepsilon}$, $\varepsilon < 0$, and $P = (\sum_j p_j^{-a})^{-1/a}$, the symmetric first-order condition is

$$\frac{p}{p-c} = \frac{a(n-1)+|\varepsilon|}{n} \equiv R, \qquad p^* = c\frac{R}{R-1}. \tag{2}$$

A finite stationary price exists exactly when $a(n-1)+|\varepsilon| > n$; otherwise the menu ceiling binds. At $(n, a, \varepsilon) = (2, 2, -0.05)$, $R = 1.025$ and $p^* = 41c$. This conditional number is a demand-model sensitivity, not an estimated market markup.

3 A history-dependent cut penalty

Suppose a provider's allocation weight is multiplied by $\theta \in [0, 1]$ if its current quote is below any of its previous M quotes. In a symmetric profile with common price q, exponent two, and rival weight $W = (n-1)q^{-2}$, a flagged provider earns

$$g(p; \theta) = \frac{\theta(p-c)}{\theta + Wp^2}.$$

The unconstrained optimum is $p_{\text{dev}} = c + \sqrt{c^2 + \theta/W}$; the best weak cut is $\widehat{p} = \min\{q, p_{\text{dev}}\}$. The calibrated value $\theta = 0.17$ comes from a conditional owned-probe selection ratio, not a randomized estimate of a proprietary rule.

The symmetric deterrence calculation is useful but insufficient. In the five-provider calibrated profile, the learned high quote survives one-step deviation checks but fails a persistent-cut audit: accepting seven penalized periods and then receiving unpenalized allocation increases discounted value by 8.17%. The learned ceiling path is therefore not an equilibrium.

3.1 Exact delayed-credit boundary

Fix rivals and restrict the subject provider to a high quote H and low quote ℓ. Let u_H be profit at H, u_L unpenalized profit at ℓ, and $u_{\theta L}$ penalized profit at ℓ. Assume

$$u_L > u_H > u_{\theta L}. \tag{3}$$

Theorem 3 (Rational memory boundary)**.** *From an all-high history, the optimal policy is either to remain high forever or to cut immediately and remain low. The cut is strictly optimal iff*

$$\gamma^M > \frac{u_H - u_{\theta L}}{u_L - u_{\theta L}}. \tag{4}$$

At equality both policies are optimal. The continuous boundary is

$$M^* = \frac{\log[(u_H - u_{\theta L})/(u_L - u_{\theta L})]}{\log \gamma}.$$

For the audited profile, $(u_H, u_{\theta L}, u_L) = (0.1067535, 0.0351280, 0.1501829)$ and $\gamma = 0.95$, so $M^* = 9.240$. Exact optimization cuts for integer $M \leq 9$ and stays high for $M \geq 10$.

Corollary 4 (Bounded-horizon wedge)**.** *When equation* (4) *holds, an open-loop receding-horizon controller with zero terminal value and horizon at most M stays high: it sees only $u_{\theta L} < u_H$. Its action differs from the infinite-horizon optimum.*

Theorem 5 (Path-equivalent temporal abstraction)**.** *Augment the primitive MDP with an option that executes $M+1$ existing low actions, while retaining both primitive actions. The augmented semi-Markov problem has exactly the same optimal value as the primitive problem at every state.*

Theorem 5 separates opportunity from discovery. The option adds no price path and no provider value. It can nevertheless change which path a bounded learning algorithm finds.

4 Calibrated environment and registered design

The deterministic request-level kernel posts quotes, samples fallback order, settles capacity and reliability, and records exact transfers. Provider species are classified on the earliest 60% of the panel using frozen rules: author-price adopters, static undercutters, active undercutters, and premium providers. Demand is an AR(1) process with intraday shape. Costs are bounded by owned-capacity and spot-dependent bands. The flow-allocation rule has no fitted parameter.

The preregistered E-SIM1 validation uses 20 seeds and a 56-epoch evaluation. Its weighted moment distance is 0.019 against a 0.04 gate. The most informative moments are not direct calibration targets: simulated flow elasticity is -0.718 ± 0.351 versus a market-conditional target -1.153; the out-of-sample author-price atom is 0.722 ± 0.082 versus 0.834; and the simulated max/min dispersion ratio is 1.991 versus 2.236. This establishes adequacy for the registered counterfactuals, not structural identification of the live router.

E-SIM5–9 use seeds 0–19, $\gamma = 0.95$, 300,000 environment transitions, and a 10,000-transition frozen-policy evaluation. Seed-bootstrap intervals quantify Monte Carlo variation under the frozen simulator. They are not confidence intervals for market parameters or calibration uncertainty.

5 Results

5.1 Falsification: state information is not enough

E-SIM5 enumerates all $2^7 = 128$ router histories. Exact value iteration cuts in all 20 profiles. A Q-learner that sees only its last action stays high in 20 of 20 seeds. Crucially, the learner that sees the complete Markov history also stays high in 19 of 20. Its median-price contrast relative to the aliased learner is -0.047, with paired interval $[-0.142, 0]$. The registered state-aliasing gate fails.

5.2 Main result: temporal abstraction closes the local gap

E-SIM6 adds a `commit-low` option equal to $M+1$ primitive low actions. Exact primitive and option values agree over every state within 10^{-10}. At $M = 7$, the option learner chooses the exact action and has at most 5% normalized regret in 18 of 20 seeds, versus 1 of 20 primitive learners. The option-minus-primitive regret contrast is -0.0643 with interval $[-0.0755, -0.0493]$.

The memory sweep gives the mechanism check in figure 1. Both learners succeed at $M \in \{1, 3, 5\}$. At $M = 7$ and 9, exact optimization still cuts while primitive learning usually fails. At $M = 12$, beyond the rational boundary, exact optimization stays high; primitive learning agrees in 20 of 20 seeds, while the option overcommits and adds 0.0942 normalized regret ($[0.0735, 0.1160]$). The intervention is helpful in the delayed-credit region, not universally.

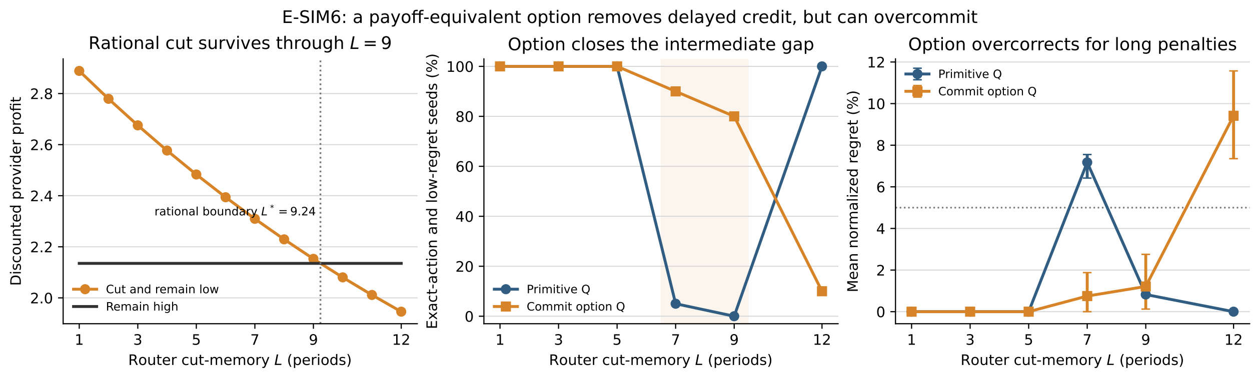

Figure 1: Exact provider value, successful learning, and normalized regret across the router memory sweep. The dotted boundary is derived from equation (4); it is not fitted to the learning outcomes.

Table 1: Registered E-SIM6 gates at the calibrated memory $M = 7$.

Gate	Threshold	Result
Exact option-value gap	$\leq 10^{-10}$	pass
Regret interval	upper endpoint < 0	pass
Option exact-action/low-regret	$\geq 16/20$	18/20
Primitive exact-action/low-regret	$\leq 4/20$	1/20
Composite mechanism gate	all components	pass

5.3 Transport and robustness

E-SIM7 applies the same design to four frozen calibrated price books. The strict transport gate fails: only two markets are delayed-credit eligible, and primitive success there is 12/20 and 14/20 rather than at most 4/20. The prespecified rational-boundary classification nevertheless aligns with all four effect signs. In eligible Kimi K2.6 and GLM-5.1 books ($M^* = 26.31, 27.91$), the option lowers normalized regret by 0.178 and 0.118. In ineligible GLM-5.2 and DeepSeek V4 books ($M^* = 2.59, 2.67$), it adds 0.155 and 0.152. This is descriptive sign transport, not confirmation of universal trap severity.

E-SIM8 varies learning rate $\alpha \in \{0.05, 0.15, 0.30\}$ and exploration decay $\beta \in \{1, 2, 4\} \times 10^{-5}$. The composite gate passes in seven of nine cells. All nine regret intervals are strictly negative, and option success is 18/20 or 19/20 throughout. The failed cells have $\alpha = 0.30$ and slower exploration decay: primitive success increases, so the severity criterion fails even though the option remains beneficial.

5.4 Falsification: an ordinary multi-step target is insufficient

E-SIM9 holds the primitive action set and every feasible price path fixed but replaces the one-step target with an eight-step Q target spanning the complete penalty window. It yields 0/20 successful learners, versus 1/20 for one-step Q and 18/20 for the option benchmark. Its normalized-regret contrast relative to one-step Q is $+0.0038$, with paired seed-bootstrap interval $[0, 0.0113]$. The registered gate fails. Thus E-SIM6 is an option-specific temporal-abstraction result, not evidence that every longer return operator solves the problem.

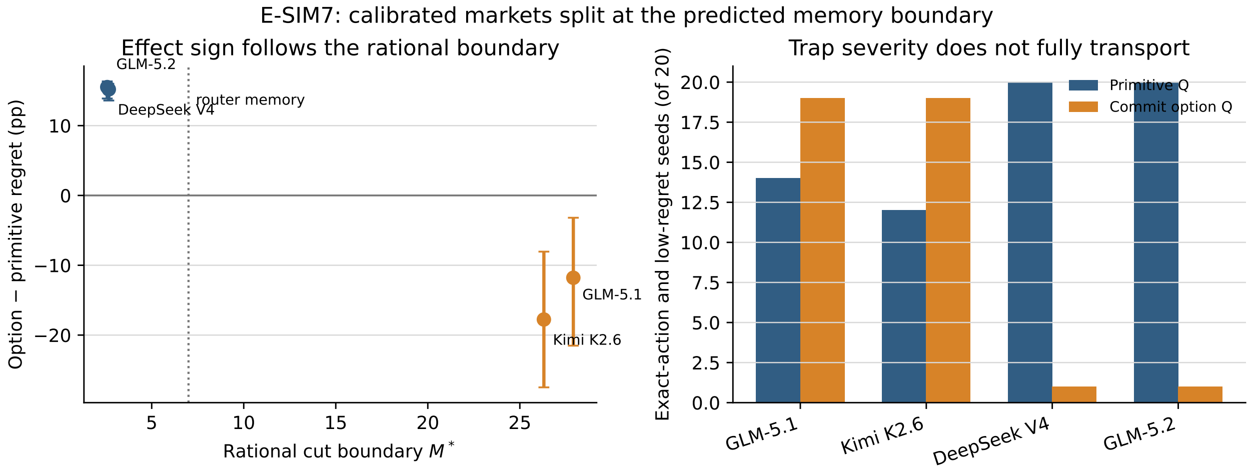

Figure 2: Cross-market transport. Effect signs align with the rational memory boundary, while the registered severe-trap gate fails.

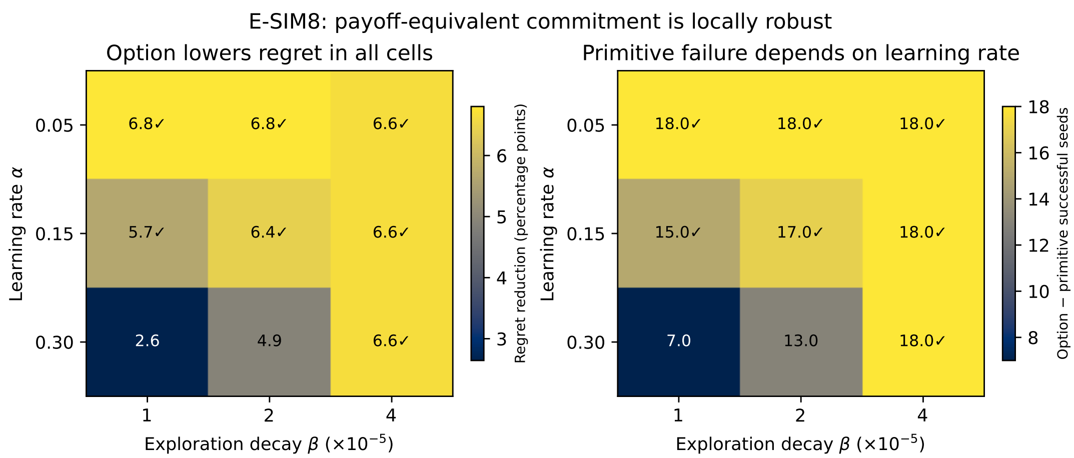

Figure 3: Local Q-learning robustness. Checkmarks denote cells satisfying the complete registered conjunction, not significance stars.

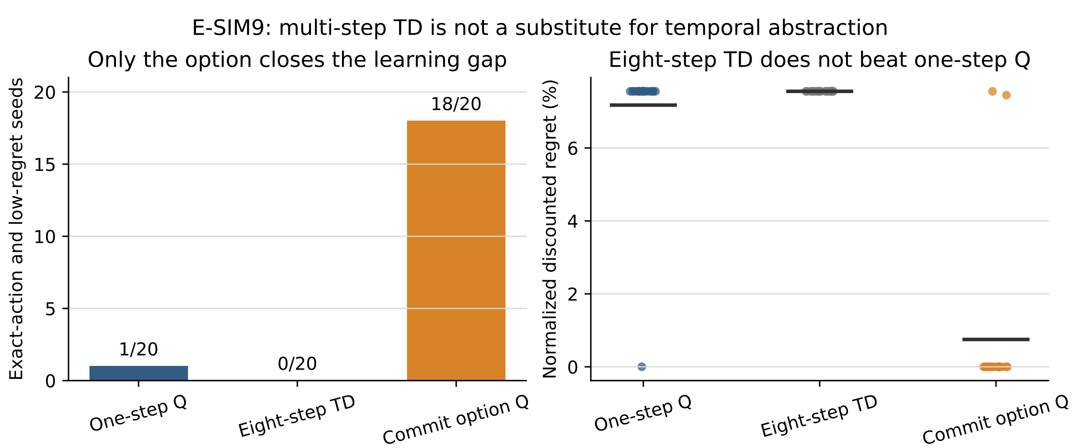

Figure 4: Registered falsification. Eight-step TD does not substitute for the path-equivalent commitment option.

6 Relation to prior work

Inverse-power allocation is a constant-elasticity differentiated-products demand system; variable-markup nested-CES models provide the closest static analogy [Atkeson and Burstein, 2008]. We treat the router exponent as a public design primitive rather than an estimated taste parameter.

Algorithmic-pricing studies show that learning protocols, timing, and platform design can alter prices [Calvano et al., 2020, Asker et al., 2022, Brown and MacKay, 2023, Johnson et al., 2023]. Brown and MacKay [2025] derive coercion when a fast firm commits to a multi-period rule that disciplines a rival. Our mechanism is different: no provider reacts to another provider's commitment rule; the router imposes an own-cut allocation penalty, and the original high-price path is rejected as an equilibrium.

Temporal abstraction and option Q-learning are classical [Sutton et al., 1999]. Recent work studies delayed reward and sequence compression directly [Han et al., 2022, Ramesh et al., 2024]. Our contribution is not the options framework. It is the derivation of the delay from a marketplace rule, the exact economic memory boundary, and the nonmonotone interaction between that boundary and a path-equivalent learning interface.

7 Limitations and claim boundary

The baseline theory has captive demand, no capacity binding, and the documented non-tool-calling price-weighted rule after eligibility filtering rather than the proprietary implementation. The cut multiplier is a conditional association from one buyer tier. Treating it as a multiplicative weight and the trailing window as exact memory is calibration, not randomized identification. Costs are identified sets.

The main experiment fixes rivals and restricts the provider to the audited high quote and best persistent cut. It identifies a finite-MDP mechanism, not an equilibrium of the full provider game. Twenty seeds vary learning randomness around one payoff profile. E-SIM8 gives local tabular-Q robustness; E-SIM7 rejects universal severe-trap transport. E-SIM9 shows that an ordinary eight-step target is not a substitute for the option; other return operators and traces remain untested. Deep RL, LLM pricing agents, endogenous rivals, executable open-source routers, and longer calibration panels are future tests. No result establishes provider intent, communication, collusion, or a causal effect in the live router.

8 Conclusion

The router is the provider's demand curve, and its memory shapes which price paths a bounded algorithm can discover. Inverse-square allocation creates a classical static markup floor. The new result is dynamic: at the calibrated seven-period cut penalty, cutting is rational but delayed credit keeps primitive Q-learning high. A path-equivalent commitment option closes that local gap and then overcorrects beyond the rational boundary. An ordinary longer TD target does not reproduce the effect, which confines the result to this action abstraction. Marketplace design should therefore evaluate both rational incentives and the learning dynamics induced by its interfaces.

A Proofs

Proof of the share and single-crossing lemma. Write $s_i = w_i/(w_i + W_{-i})$, where $w_i = p_i^{-a}$ and W_{-i} is fixed. Differentiation gives $\partial s_i/\partial p_i = -(a/p_i)s_i(1 - s_i)$. Hence

$$\frac{\partial \pi_i}{\partial p_i} = Ds_i \left[1 - a(1 - s_i)\frac{p_i - c_i}{p_i}\right].$$

Both $1 - s_i(p_i)$ and $(p_i - c_i)/p_i$ are strictly increasing above cost. Their product crosses one at most once, which proves quasiconcavity and the best-response statement. □

Proof of theorem 2. At symmetry $s_i = 1/n$. The interior first-order condition becomes $a(1 - 1/n)(p - c)/p = 1$, which yields the formula and requires $a(n - 1) > n$. If the inequality fails, the derivative at every symmetric interior profile is positive and the cap is the only symmetric solution. At any interior equilibrium, $(p_i - c_i)/p_i = 1/[a(1 - s_i)] > 1/a$, which rearranges to the markup floor. As $s_i \to 0$, the inequality converges to equality. □

Proof of theorem 3. In the all-low state, low strictly dominates high by equation (3); a high action would also reinsert a high quote into memory. Before the all-low state, any high action resets progress. Any policy that eventually cuts can therefore be weakly improved to k high actions followed by low forever. Its value is

$$V_k = \frac{u_H(1 - \gamma^k)}{1 - \gamma} + \gamma^k V_{\text{cut}}, \quad V_{\text{cut}} = \frac{(1 - \gamma^M)u_{\theta L} + \gamma^M u_L}{1 - \gamma}.$$

Since $V_k - u_H/(1 - \gamma) = \gamma^k[V_{\text{cut}} - u_H/(1 - \gamma)]$, either $k = 0$ or never cutting is optimal. Comparing these alternatives and rearranging gives equation (4). □

Proof of theorem 5. The augmented action set contains all primitive actions, so its optimal value is weakly larger. Conversely, replace every option in any augmented policy by its finite defining sequence of primitive actions. The unrolled policy is feasible in the primitive MDP and produces the same states and discounted rewards. Its value is therefore identical, so the augmented optimum cannot be larger. □

B Reproducibility

The clean confirmatory bundles are E-SIM5 `3e5a55405e`, E-SIM6 `93265b9b03`, E-SIM7 `bd74ab2eb7`, and E-SIM8 `4d84b9b3a2`; the registered E-SIM9 falsification is `179eca0f9d`. Every manifest records source commit `4f9007b`, a result hash, input-artifact hashes, and equality between the executed market-environment source and the source stored at that commit. Exact Bellman residuals, option-value equality, and paper-number crosswalks are enforced by the test suite.

References

John Asker, Chaim Fershtman, and Ariel Pakes. Artificial intelligence, algorithm design, and pricing. *AEA Papers and Proceedings*, 112:452–456, 2022. doi: 10.1257/pandp.20221058.

Andrew Atkeson and Ariel Burstein. Pricing-to-market, trade costs, and international relative prices. *American Economic Review*, 98(5):1998–2031, 2008. doi: 10.1257/aer.98.5.1998.

Zach Y. Brown and Alexander MacKay. Competition in pricing algorithms. *American Economic Journal: Microeconomics*, 15(2):109–156, 2023. doi: 10.1257/mic.20210158.

Zach Y. Brown and Alexander MacKay. Algorithmic coercion with faster pricing. Technical Report 34070, National Bureau of Economic Research, 2025.

Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, 110(10):3267–3297, 2020. doi: 10.1257/aer.20190623.

Beining Han, Zhizhou Ren, Zuofan Wu, Yuan Zhou, and Jian Peng. Off-policy reinforcement learning with delayed rewards. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 8280–8303, 2022.

Justin P. Johnson, Andrew Rhodes, and Matthijs R. Wildenbeest. Platform design when sellers use pricing algorithms. *Econometrica*, 91(5):1841–1879, 2023. doi: 10.3982/ECTA19978.

OpenRouter. Provider routing: Intelligent multi-provider request routing. `https://openrouter.ai/docs/guides/routing/provider-selection`, 2026. Accessed 2026-07-18.

Aditya Ramesh, Kenny John Young, Louis Kirsch, and Jürgen Schmidhuber. Sequence compression speeds up credit assignment in reinforcement learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 42104–42126, 2024.

Richard S. Sutton, Doina Precup, and Satinder Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211, 1999. doi: 10.1016/S0004-3702(99)00052-1.